

Najma Akter Lopa

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 Najma-Lopa |  Najma-Lopa

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Betura, Biral, Dinajpur, Bangladesh

OBJECTIVE

To build an academic career as a Lecturer in Computer Science and Engineering by contributing to quality teaching, student supervision, and research. My interests include Machine Learning, Explainable AI, Artificial Intelligence, and Software Development, supported by research publications, academic projects, and hands-on development experience. I aim to foster students' technical and analytical abilities while pursuing impactful research and contributing meaningfully to the academic community.

EDUCATION

- **Hajee Mohammad Danesh Science and Technology University** March, 2022 – August, 2026
B.Sc. in Computer Science and Engineering
◦ CGPA : 3.82/4.00 (Ranked 4th in Batch) Dinajpur, Bangladesh
- **Dinajpur Government College** 2018 – 2020
Higher Secondary Certificate (Science) Dinajpur, Bangladesh
◦ GPA: 5.00/5.00
- **Betura B.L High School** 2016 – 2018
Secondary School Certificate (Science) Dinajpur, Bangladesh
◦ GPA: 5.00/5.00

RESEARCH WORK

Publications: C=Conference, U=Under Review, E=Extended Abstract, D=Dataset Contribution

- [U.1] **Najma Akter Lopa**, Sultan Mahamud Opu, Md. Tanvir Ahmmmed Shuvo and Pankaj Bhowmik, "Towards Responsible Clinical AI: An XAI-Driven Framework for Cardiovascular Disease Risk Assessment with Clinical Decision Support", 29th International Conference on Computer and Information Technology (ICCIT 2026), IEEE
- [C.1] Sourav Roy, **Najma Akter Lopa**, Sultan Mahamud Opu, Shafiqul Islam Sazu, Md. Ridwanur Rahman Shuvo, Raghubir Chaudhary, Ashis Kumar Mandal and Pankaj Bhowmik, "Explainable Machine Learning Framework for Early Detection of Depression in University Students", 2026 IEEE International Conference on Engineering and Frontier Technologies (ICEFronT 2026), IEEE (Accepted)
- [D.1] **Najma Akter Lopa**, Sultan Mahamud Opu, Sourav Roy, Mst Ifat Zahan Soma and Pankaj Bhowmik (2026), "A Cross-Sectional Dataset of Sleep Disorder Indicators Among Bangladeshi University Students", [Thesis-based Primary Dataset], Mendeley Data, V1. DOI: [10.17632/9djgvtj8d.1](https://doi.org/10.17632/9djgvtj8d.1).
- [C.2] **Najma Akter Lopa**, Sultan Mahamud Opu, Sourav Roy, Mst. Ifat Zahan Soma, and Pankaj Bhowmik (2026). "AI-Driven Analysis of Sleep Disorders in Bangladeshi University Students Using Clinical and Behavioral Insights". In 5th IEEE International Conference on Electrical, Computer & Telecommunication Engineering (ICECTE 2026), [Thesis-based Conference], DOI: [10.1109/ICECTE69292.2026.11429299](https://doi.org/10.1109/ICECTE69292.2026.11429299)
- [E.1] Sourav Roy, **Najma Akter Lopa**, Sultan Mahamud Opu, Shafiqul Islam Sazu, Md. Ridwanur Rahman Shuvo, Raghubir Chaudhary, Ashis Kumar Mandal and Pankaj Bhowmik, "Enhancing Early Detection of Depression in Academic Settings: An XAI-Guided Machine Learning Approach for Bangladeshi University Students". In Scholars' Summit 2025, HSTU Research Society. [Accepted], [Certificate](#)
- [E.2] **Najma Akter Lopa**, Sultan Mahamud Opu, Sourav Roy and Pankaj Bhowmik, "Towards an AI-Driven Framework for Effective Sleep Disorder Prediction Analyzing Clinical and Behavioral Insights", RUEC 1st International Research Conference (2025). [Accepted], [Certificate](#)

Thesis: D=Defense Completed, W=Working on it

- [D.1] **Najma Akter Lopa**, Sultan Mahamud Opu, Sourav Roy (2026). "Sleep Disorder Risk Assessment and AI-Based Clinical Decision Support Among University Students". In **Level 4 Semester II** of B.Sc. (Engineering) in Computer Science and Engineering, supervised by Pankaj Bhowmik and Dr. Masud Ibn Afjal.

[D.2] Najma Akter Lopa, Sultan Mahamud Opu, Sourav Roy (2025). "A Comparative Analysis of Gender Disparity in Programming Skills Using Educational Data Mining". In **Level 4 Semester I** of B.Sc. (Engineering) in Computer Science and Engineering, supervised by Pankaj Bhowmik and Dr. Masud Ibn Afjal.

PROJECTS

- **Depression Diagnostic System**
Tools: Python, Scikit-learn, CatBoost, SHAP, LIME, Streamlit

 - Developed an explainable ML-based depression severity prediction system.
 - Applied Bayesian optimization, stress testing, ROC-AUC, and 5-fold cross-validation; achieved 98.12% accuracy with a stacked ensemble.
 - Integrated SHAP/LIME explanations and deployed the system using Streamlit.

2026
[Live Demo](#)
- **HSTU Student Complaint Management System**
Tools: HTML, CSS, JavaScript, Backend, MySQL

 - Developed a web-based complaint management system with unique complaint ID tracking.
 - Implemented real-time complaint status updates and student feedback.
 - Developed an admin panel for complaint review, status management, and monitoring.

June 2026 – August 2026
[Live Demo](#)
- **Personal Local Cloud Storage and File Sharing System**
Tools: Python, FastAPI, MySQL, HTML, CSS, JavaScript, Ngrok

 - Developed a local cloud platform for secure file upload, storage, organization, and sharing.
 - Implemented QR/IP-based access, user dashboards, folder management, and file recovery.
 - Added encrypted storage and Ngrok-based external file sharing.

May 2026 – July 2026
[Personal Cloud](#)
- **HSTU_S&T_Hub: Real-Time Communication System**
Tools: Flutter, Supabase, Firebase

 - Enabled teachers and Class Representatives (CRs) to manage schedules, rooms, and academic communication.
 - Implemented real-time messaging, role-based access, and an AI chatbot using Firebase and Supabase.

January 2025 – June 2025
[HSTU_S&T_Hub](#)
- **World Clock and Time Zone Converter**
Tools: Flutter, Dart, Time Zone API

 - Developed a cross-platform Flutter app for real-time clock and time zone conversion.
 - Implemented dynamic time zone comparison and real-time clock functionality.
 - Added dark/light themes and responsive UI for web and mobile platforms.

November, 2024
[Live Demo](#)
- **HSTU Classroom Allocation System**
Tools: Java, JavaFX, MySQL

 - Implemented time-slot-based classroom allocation with conflict prevention using JavaFX and MySQL.
 - Developed a responsive desktop UI to manage departments, rooms, and real-time allocation data.

August 2024 – November 2024
[HSTU Classroom Allocation System](#)
- **IT Job Guide Line**
Tools: Java, JavaFX

 - Developed an interactive job guidance system to help students explore career paths, skills, and opportunities.
 - Implemented personalized recommendations, resource management, and user-friendly JavaFX interfaces.

January 2024 – June 2024
[IT Job Guide Line](#)

EXPERIENCE

- **IEEE Computer Society HSTU Student Branch Chapter**
Publicity Coordinator

March, 2026 – September, 2026
- **Programmers Arena (HSTU)**
Office Secretary

September, 2025 – Present
- **Programmers Arena (HSTU)**
Publication and IT Secretary

January, 2025 – September, 2025
- **Programmers Arena (HSTU)**
Executive Committee Member

February, 2024 – January, 2025
- **HSTU Research Society (HSTURS)**
Executive Committee Member

2023 – 2024

ACHIEVEMENTS AND AWARDS

- **Honorable Dean's Award (3×)** 2023, 2024, 2025-2026
 - Received the Honorable Dean's Award three times from the Faculty of Computer Science and Engineering, HSTU.
- **Runner-up – Scholars' Summit 2025** November, 2025
 - *Enhancing Early Detection of Depression in Academic Settings: An XAI-Guided Machine Learning Approach for Bangladeshi University Students*, Segment A (Science & Engineering), HSTU Research Society.
- **4th Place – Rangpur Divisional Project Showcasing** June, 2025
 - *HSTU.T&S.HUB: A Real-Time Communication System.*
- **11th position – Rangpur Divisional Collegiate Programming Contest at HSTU** June, 2025
- **Participant – KUET Inter University Programming Contest (IUPC)** January, 2025
- **12th position – Rangpur Divisional Collegiate Programming Contest at BRUR** October, 2024

SKILLS

- **Languages:** Strong reading, writing, and speaking competencies in Bangla and English.
- **Programming Languages:** C, C++, Java, Python, Dart.
- **Web Technologies:** HTML5, CSS3, FastAPI.
- **Frameworks:** Flutter, JavaFX, Streamlit.
- **Database Systems:** MySQL, Firebase, Supabase.
- **Data Science & Machine Learning:** Pandas, NumPy, Matplotlib, Seaborn, Scikit-learn, PyTorch, SHAP, LIME.
- **Machine Learning:** Classification, Regression, Ensemble Learning, Feature Engineering, Feature Selection, Hyperparameter Optimization, Cross-Validation.
- **Specialized Area:** Explainable AI (XAI), Stacking Ensemble, Bayesian Optimization, Statistical Analysis, ML Model Evaluation.
- **Deep Learning & NLP:** Neural Networks, CNN, RNN, LSTM, Transformers, Natural Language Processing, Text Classification, Text Preprocessing.

REFERENCES

1. **Pankaj Bhowmik**
Assistant Professor, Department of Computer Science and Engineering
Hajee Mohammad Danesh Science & Technology University, Dinajpur
Email: pankaj.cshstu@gmail.com
Phone: +8801791848439
Relationship: [e.g., Thesis Supervisor, Research Advisor & Research Collaborator etc.]
2. **Dr. Masud Ibn Afjal**
Professor, Department of Computer Science and Engineering
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Relationship: [e.g., Research Advisor, Thesis Co-Supervisor, etc.]

DECLARATION

I hereby declare that all the information provided above is true and correct to the best of my knowledge and belief.